

ML signal discovery from SecDB: a scoping map for a 10-12 week internship

Executive summary

The single highest-value project angle for a GS quant-dev intern with SecDB access is to turn risk-system outputs — VaR, scenario P&L, factor risk decompositions, aggregated book Greeks — into *alpha signals* rather than use them only as risk controls. Intermediary asset pricing theory (He-Krishnamurthy 2013, Adrian-Etula-Muir 2014, He-Kelly-Manela 2017) and the gamma-hedging intraday-momentum literature (Baltussen-Da-Lammers-Martens 2021, Barbon-Buraschi 2021) already show that dealer-balance-sheet and dealer-gamma variables price returns across asset classes. External researchers have to reconstruct these from stale Fed Z.1 tables and open-interest heuristics; SecDB gives the real thing, daily, across every product, with the correct dealer sign.

The second differentiator is curve-fitting residuals. SVI, SABR and Nelson-Siegel residuals are *proprietary micro-liquidity and demand-pressure signals* generated as a byproduct of calibration (Gârleanu-Pedersen-Poteshman 2009 give the theory). External researchers cannot reproduce them at scale.

Methodologically, gradient-boosted trees (LightGBM/XGBoost) on well-engineered tabular features are the default workhorse and win the great majority of published horse-races and essentially every finance Kaggle competition. Deep learning for time series is largely oversold — DLinear and N-HiTS beat Informer/Autoformer on almost every benchmark (Zeng et al. 2022), [Liner](#) and the M5 top-50 is ~95% LightGBM. Deep nets have a real edge only in limit-order-book microstructure (DeepLOB), text (FinBERT), representation learning (autoencoders as feature extractors), and derivatives hedging (Buehler deep hedging). RL for alpha is not recommended; RL for execution and hedging is defensible but tight on 12 weeks.

The validation problem dwarfs the modelling problem. Bailey-Borwein-López de Prado (2014) show that with 5 years of data you can afford ~45 independent trials before the expected best-in-sample Sharpe of 1.0 implies zero true skill. A 12-week project that reports a headline Sharpe without purged/embargoed CV, deflated Sharpe, transaction costs, and performance decomposed by regime is not a credible project — regardless of the model. Plan to spend ~25% of time on validation infrastructure before touching the "ML" part.

Presentation to the trading floor works best when framed as a causal hypothesis ML was used to test, not as "my black box found alpha." Lead with Information Coefficient, Sharpe, turnover, drawdown and capacity; interpret features with SHAP and one chart per claim; always include a regularized-linear baseline so the audience can see what the ML adds.

Part 1 — Landscape survey

A. SecDB, Slang and analogous systems

SecDB is a 30+ year-old in-memory object store (SecServ) organized into eventually-consistent replication groups called *rings* synchronized by SecSync. The Goldman developer blog (Oct 2023) documents ">10,000 globally distributed custom object databases, 2.5 billion connections, 164 TB of messages, and 8 PB of data served." Every SecDB object is **bi-temporal** (value-time and transaction-time) and sits in a **dependency graph** that lazily invalidates and recomputes prices, Greeks and scenarios when any market input changes. This is the machinery behind the firm's "one price" principle.

Slang is a proprietary C-like interpreted DSL created in 1992 with tight SecDB integration. Per Michal Dorko's InfoQ talk, the code base is **>200M lines, ~3,000 developers over 30 years, ~160M daily jobs, >300M compute-hours per week**, currently moving from tree-walker interpreter to bytecode+JIT. The system was built by Dubno/Gribble/Lundeen under Armen Avanesians in J. Aron; Martin Chavez drove firmwide adoption. (The Block) (The task prompt's reference to "Armstrong/Korn" does not match any public source and is likely a name confusion — the intern should confirm with their sponsor.)

Data you can realistically pull as time series for ML features: curves (daily and often intraday), vol surfaces, fixings, spots, forwards, implied-vol node histories; book-level Greeks (delta/gamma/vega/theta + cross-Greeks) from nightly risk cubes; scenario P&Ls at standard shock nodes; factor VaR and component VaR decompositions; trade populations. Entitlements are desk-scoped; writing to SecDB typically needs senior sign-off, but read-only sandboxes are standard.

Analogous systems all share the SecDB architectural DNA because the same people built them in sequence:

- **JPMorgan Athena** — Python-based, built by Kirat Singh and Mark Higgins (both ex-GS SecDB). (Medium) PyData London 2018 documented 35M LOC Python, 150k modules, 4,500 devs, 20k commits/week. (Hacker News) (YouTube) JPM's OmniAI (internal SageMaker wrapper) (Disabled Person) and Athena Jupyter Platform expose this to research and ML workflows. (WORK180)
- **Bank of America Quartz** — Python, ~17M LOC, led by Dubno (then CIO of Global Markets Tech) and Singh again post-Merrill.
- **Beacon Platform** — the commercial cloud-native heir founded by Singh+Higgins in 2014; (LinkedIn) (APAC Entrepreneur) the closest publicly documented description of SecDB-class architecture.
- **Morgan Stanley, Deutsche Bank** — no equivalent public architectural documentation; treat as proprietary.

GS Quant + Marquee are the public projection. The open-source (gs-quant) Python SDK (~9.9k GitHub stars, Apache 2.0) exposes a subset of internal pricing models and risk measures (GitHub +2) with (PricingContext), (HistoricalPricingContext), (Portfolio), (GitHub) 400+ datasets via (Dataset)/(DataApi), a backtester, and REST API wrappers. The firm-facing docs

(developer.gs.com) describe the Marquee data surface. Using ([gs-quant](#)) externally is the single best way to practice SecDB-style workflows before the internship starts.

B. ML methods that actually work for signal discovery

The evidence-weighted hierarchy is unambiguous:

Tier 1 — the workhorses the intern should default to. Gradient-boosted trees (LightGBM, XGBoost, CatBoost) win Gu-Kelly-Xiu (2020 RFS, the canonical 60-year US equity horse-race), ([Oxford Academic](#)) the Optiver Realized Vol Kaggle, Jane Street, Ubiquant, and 95% of the M5 top-50. They handle mixed-type tabular features, missing data, monotonic constraints, asymmetric losses, and retrain in minutes. Failure modes: they extrapolate as constants (bad in regime shifts), overfit fast on low-SNR data without aggressive regularization, and SHAP importance is unstable under correlated features. **Regularized linear models (Ridge, Elastic Net, Lasso)** are the mandatory baseline — Kozak-Nagel-Santosh (2020 JFE) show that ridge-shrunk SDFs on PCs of characteristic portfolios match or beat non-linear ML, ([Oxford Academic](#)) and Chinco-Clark-Joseph-Ye (2019 JF) show Lasso wins at 1-minute horizons. ([arXiv](#)) If your GBM doesn't beat a ridge on the same features, you haven't learned anything.

Tier 2 — scoped additions with real evidence. Autoencoders work as *upstream feature extractors*, not as stand-alone predictors: Gu-Kelly-Xiu (2021 JFE) "Autoencoder Asset Pricing Models" ([Substack](#)) delivers cleaner out-of-sample pricing errors than IPCA or Fama-French ([SSRN](#)) *when embedded in a no-arbitrage factor structure*. Factor-model + ML-on-residuals is arguably **the single best design pattern for a SecDB intern**: decompose returns with Barra-style factors, ML on ϵ , reconstruct factor-neutral portfolios. Gaussian processes are competitive for arbitrage-constrained vol-surface smoothing (Cousin et al. 2022, Ackerer et al. NeurIPS 2020) but not for return prediction.

Tier 3 — hype zones. Deep learning for time series is mostly oversold. Zeng et al. (AAAI 2023) "Are Transformers Effective for Time Series Forecasting?" shows a one-layer linear model (DLinear) ([Datasciencewithmarco](#)) beats Informer/Autoformer/FEDformer on 9 of 9 benchmarks. ([AAAI](#)) If the intern attempts sequence DL, use PatchTST (Nie et al. ICLR 2023), ([GitHub +2](#)) N-HiTS or ModernTCN — not the Informer family. GNNs on cross-asset relationships (Feng et al. 2019, HATS 2019) ([arXiv](#)) ([arXiv](#)) have headline backtests that almost universally contain leakage via full-sample correlation graphs; no tier-1 firm publicly endorses them for production. RL for *execution* (Nevmyvaka-Feng-Kearns 2006) and *derivatives hedging* (Buehler et al. 2019 Deep Hedging) ([Taylor & Francis Online +2](#)) is defensible; RL for *alpha or portfolio allocation* (FinRL family) should not be cited as evidence that anything works.

Horizon × asset-class defaults: daily single-name equities → GBM on characteristics, ridge baseline; intraday order-book → TCN/LSTM/attention with GBM baseline; options/vol surfaces → arbitrage-constrained GP or NN smoothing + GBM on implied-minus-realized spreads; cross-asset monthly → autoencoder-latent-factors + GBM, or factor-residual GBM; credit monthly → GBM on spread features with meta-labeling for sparse defaults.

This is where a SecDB-authored project can genuinely differentiate. The features below are organized from best-established to most novel.

Greeks aggregated as dealer-positioning features. Barbon-Buraschi (2021 SSRN 3725454) "Gamma Fragility" show that stock-level gamma imbalance interacted with illiquidity drives intraday momentum (when dealers are net short gamma) versus mean reversion (when net long). Ni-Pearson-Potesman (RFS 2021 and JFE 2005) document pinning at strikes on expiry driven by dealer delta-rehedging as gamma explodes. (American Finance Associat...) The practical construction — **aggregate dealer GEX, per-ticker gamma imbalance, vanna exposure (drives FOMC/CPI-day response), charm exposure (end-of-day drift), volga (vol-of-vol sensitivity), speed/color (curvature of hedging need), gamma-flip level (regime boundary), distance-to-flip** — is standard, but *public* GEX assumes dealers are short calls and long puts about 70% correctly (Squeezemetrics 2018); SecDB has the true book sign. This is a clean competitive edge.

Vol surface features with published cross-sectional evidence. The 25-delta volatility smirk (Xing-Zhang-Zhao 2010 JFQA) predicts single-stock returns; (ScienceDirect) steepest-smirk decile underperforms flattest by ~10.9%/yr for six months. (Semantic Scholar) The put-call IV spread (Cremers-Weinbaum 2010 JFQA) predicts +50 bps/week. (Cambridge Core) The variance risk premium $IV^2 - RV^2$ (Bollerslev-Tauchen-Zhou 2009 RFS) explains >15% of quarterly SPX returns. (Federal Reserve) (SSRN) Risk-neutral kurtosis positively predicts returns (Bali-Hu-Murray); risk-neutral skewness is contested (Conrad-Dittmar-Ghysels negative; Stilger-Kostakis-Poon positive — term-horizon-dependent per Feunou et al.). **Critical caveat: Muravyev-Pearson-Pollet (2022) argue smirk and IVS partly proxy for stock-borrow fees;** (ResearchGate) **any ML model using them must control for short interest and borrow cost.** Surface PCA (Cont-da Fonseca 2002, Avellaneda et al. 2020) compresses daily IV-surface deltas into ~3 factors (SSRN) (level, skew, term); loadings on these factors are clean, low-cardinality, stationary features. VVIX and single-stock vol-of-vol are priced in the cross-section (Baltussen-van Bakkum-van der Grint 2018 JFQA: ~-10%/yr premium for high-VOV stocks). (Semantic Scholar)

Term-structure and curve features. Nelson-Siegel level/slope/curvature (Diebold-Li 2006) predict bond, equity and REIT one-month returns; (MDPI) 2s10s slope predicts recession probability and equity drawdowns. The **cross-currency basis** (Du-Tepper-Verdelhan 2018 JF) widens at quarter-ends on G-SIB balance-sheet constraints (Bank for International Settl...) and is a direct proxy for dealer balance-sheet stress; CDS-bond residual basis (Bai-Collin-Dufresne 2019) negatively predicts bond returns; (ScienceDirect) repo-GC spreads and swap spreads are classical fire-sale indicators; CDS-curve inversions signal distress (Han-Zhou).

Correlation and factor-residual features. Top eigenvalue share $\lambda_1/\Sigma\lambda$ of the rolling correlation matrix (Bouchaud-Potters random-matrix-theory framework) (Cfm) is a "correlation regime" signal that rises into drawdowns. Residuals ε_i from Barra-style factor decompositions drive the Avellaneda-Lee (2010) statistical-arbitrage literature — 1-to-5-day mean reversion on idiosyncratic returns. Kritzman-Li-Page-Rigobon (2011 FAJ) "Absorption Ratio" (share of variance captured by top N PCs) predicts drawdowns as a systemic-risk indicator.

Risk-system outputs as predictive features (the novel angle). The theoretical backing is rigorous: He-Krishnamurthy (2013 AER) prove that risk premia rise non-linearly when intermediary equity-capital constraints bind; (American Economic Associ...) (Zhiguohe) Adrian-Etula-Muir (2014 JF) show a single-factor intermediary-leverage SDF (IDEAS/RePEc) delivers $R^2=77\%$ on 25 size/BM + 10 momentum + 6 bond portfolios; (IDEAS/RePEc) He-Kelly-Manela (2017 JFE) extend this to equity, bonds, FX, options, CDS, commodities with a single pricing kernel; (IDEAS/RePEc) Adrian-Shin (2010 JFI) show dealer repos forecast VIX innovations. (SSRN +2) Every one of these papers uses *crude public proxies* — quarterly Flow-of-Funds, Compustat leverage. **SecDB can produce daily, dealer-internal, cross-asset versions:** Δ firm VaR, component VaR by asset class, incremental VaR spikes, factor-VaR concentration (Herfindahl) as a crowding indicator, rolling rank of which stress scenario is currently worst, scenario-P&L dispersion (low dispersion = hidden concentration), VaR-utilization % as a forced-deleveraging proxy. Combined with Coval-Stafford (2007 JFE) fire-sale evidence and Adrian-Brunnermeier (2016 AER) CoVaR methodology, this is a research programme, not a single feature.

Curve-fitting residuals as demand-pressure signals. Gârleanu-Pedersen-Poteshman (2009 RFS) "Demand-Based Option Pricing" proves that end-user option demand deviations persistently distort IVs from the no-arbitrage price. The practical consequence: **residuals from SVI, SABR and Nelson-Siegel calibrations are not noise — they contain demand/flow information and predict mean-reversion.** Store them daily as (date $\times$ strike $\times$ expiry) tensors, apply functional PCA, and PC1 becomes a "demand pressure" feature, PC2 a "term-structure distortion" feature. No external researcher can replicate SecDB's per-strike residual history at scale.

D. The graveyard — what fails and why

Failure modes, ranked by how often they destroy junior projects: data-snooping / multiple hypothesis testing (Harvey-Liu-Zhu 2016 RFS argue the $t>2$ bar is too low — a new factor needs $t>3$ (Oxford Academic) given the 316+ already tested); (OpenSourceQuant) overfitting and p-hacking (Bailey-Borwein-López de Prado 2014 prove that with 5 years of data, trying >45 independent configurations almost guarantees in-sample Sharpe of 1 with OOS Sharpe of 0); lookahead bias via point-in-time violations (restated risk figures, delayed fundamentals); survivorship bias on current index constituents; non-stationarity; ignoring transaction costs and financing; overfitting the *label* definition; feature-importance instability across folds.

The López de Prado framework is the practitioner standard. From *Advances in Financial Machine Learning* (2018): (Hudson & Thames) dollar bars beat calendar bars; triple-barrier labelling (PhilPapers) (take-profit, stop-loss, expiration) (Reasonabledeviations) with volatility-scaled thresholds; (Substack) sample weights by average uniqueness to break overlap-induced IID violations; (StudyLib) fractional differentiation (pick the smallest d that passes ADF) to make price features stationary without losing memory; **purged K-fold CV with embargo** (Wikipedia) (StudyLib) as baseline validation; (ScienceDirect) **Combinatorial Purged Cross-Validation (CPCV)** (StudyLib) to produce a *distribution* of Sharpes from a single history; (ScienceDirect) MDI/MDA/SFI feature importance (MDA across folds is the single best stability check); meta-labelling (primary model for *side*, secondary binary for *size*) (Hudson & Thames +2) to convert a low-precision primary into a filtered ensemble. *Machine Learning for Asset Managers* (2020) adds the False Strategy

Theorem — with 1,000 skill-less strategies, $E[\text{max Sharpe}] \approx 3.26$. (GitHub) Known critiques: AFML examples are synthetic; meta-labelling itself can overfit if not nested-CV'd; the (ONC) implementation has had bugs (GitHub) (use mlfinlab fork). (GitHub)

The Bailey-Borwein-López de Prado statistical toolkit is non-negotiable: **Deflated Sharpe Ratio** adjusts for number of trials, variance of trial Sharpes, sample length, skewness and kurtosis; (ScienceDirect +3) **Probability of Backtest Overfitting (PBO)** via Combinatorially Symmetric CV (Western Michigan University) tells you whether your selection process is itself overfitting (Glama) ($\text{PBO} > 0.5 = \text{broken}$); Minimum Backtest Length formula gives hard upper bounds on how many configs you can honestly try. (StarQube)

Harvey-Liu (2015 JPM) haircut Sharpe methodology — available as Matlab/R code at people.duke.edu/~charvey/backtesting — applies Bonferroni/Holm/BHY-FDR multiple-testing corrections to a reported Sharpe. (Rdrr.io) (GitHub) The industry rule-of-thumb "50% haircut" is wrong; marginal Sharpes get haircut much more; exceptional ones barely at all. (Rdrr.io)

How practitioners at top firms validate. Renaissance publishes nothing. Citadel and DE Shaw are nearly silent. Two Sigma publishes methodology but not live signals. Public consensus across AQR (Asness-Chandra-Ilmanen-Israel 2017 on factor timing: "if you sin, sin only a little"), Man AHL/Group (Harvey et al. 2019 "The Best of Strategies for the Worst of Times"), Robert Carver (idea-first not data-first; explicit/implicit/tacit fitting all dangerous), (Slideshare) and Ernie Chan (require Sharpe > 1 and max-DD duration $< \sim 1$ year; paper trading as the only true OOS) converges on: hypothesis before data, purged-CV + walk-forward + DSR + haircut, transaction costs always included, adversarial peer review, small-capital incubation for 6-12 months before scaling.

E. Cross-asset and macro signal research (heavy)

Options → equity is the richest documented direction: IV smirk predicts returns (Xing-Zhang-Zhao), (ScienceDirect) IV spread predicts returns (Cremers-Weinbaum), (Cambridge Core) informed option volume predicts returns (Pan-Potesman 2006 RFS), (IDEAS/RePEc) (Oxford Academic) variance risk premium predicts market (Bollerslev-Tauchen-Zhou), (Federal Reserve) short-sale-constrained parity violations predict returns (Ofek-Richardson-Whitelaw 2004). Caveat: effects have decayed post-2005 and are partially explained by borrow fees.

Rates ↔ FX is dominated by the carry literature (Lustig-Roussanov-Verdelhan 2011, Koijen-Moskowitz-Pedersen-Vrugt 2018 unified carry across nine asset classes (CBS Research Portal) with combined Sharpe ~ 0.9) and the cross-currency basis literature (Du-Tepper-Verdelhan 2018 on CIP deviations (American Economic Associ...) as dealer-constraint proxies). (NBER +2) Brunnermeier-Nagel-Pedersen (2008) show FX risk-reversal skew prices carry-crash risk.

Credit → equity: Acharya-Johnson (2007 JFE) find CDS spread changes negatively predict equity returns (ScienceDirect) *around negative credit events* for firms with many banking relationships (informed-lender channel); (ScienceDirect) Hilscher-Pollet-Wilson (2015 JFQA) show equity leads CDS unconditionally and CDS leads only conditionally. (ScienceDirect) CDS-bond basis and Kapadia-Pu-style cross-market option-CDS deviations mean-revert on ~ 1 -week horizons.

Commodities and macro: Gorton-Hayashi-Rouwenhorst (2012) show commodity-futures basis (curve slope) strongly predicts returns via theory-of-storage; Etula (2013) bridges commodities to intermediary-leverage. The copper-gold ratio is practitioner folklore and has broken since 2021 — do not use as a hard signal.

Cross-asset factor models with ML — Gu-Kelly-Xiu (2020 RFS) remains the canonical horse-race; (Oxford Academic) IPCA (Kelly-Pruitt-Su 2019 JFE) extended to options (Büchner-Kelly 2022) and bonds (Kelly-Palhares-Pruitt 2023) provides a unified cross-asset framework; Freyberger-Neuhierl-Weber (2020 RFS) show adaptive group LASSO with nonparametric spline terms gets ~50% Sharpe uplift over linear; (NBER) (Becker Friedman Institute) Chen-Pelger-Zhu (2024 Management Science) use a no-arbitrage-constrained GAN+LSTM. (arXiv) (Stanford University)

Risk-system outputs as alpha, operationalized: Baltussen-Da-Lammers-Martens (2021 JFE) "Hedging Demand and Market Intraday Momentum" is the direct bridge — aggregate dealer net gamma determines whether the last 30 minutes amplify or dampen the day's return, across 60+ futures globally. Replicating and extending this in SecDB using real firm Greeks across non-equity-index markets is the single most concrete project design the intern could pitch. Combined with Adrian-Shin (dealer leverage forecasts VIX), Adrian-Brunnermeier CoVaR, Kritzman et al. Absorption Ratio, and the intermediary-leverage factor (He-Kelly-Manela), this is a complete research programme.

F. Regime detection and non-stationarity

Hidden Markov Models (2-3 states) on returns+volatility are the industry default — cheap (two days with (hmmlearn)), interpretable, but limited: Gaussian HMMs assume normal returns, Markov property ignores long memory, and critically, **HMMs confirm regime changes rather than predicting them**. Random Forests on macro features often outperform HMMs for *next-period* regime classification.

Change-point detection: CUSUM (Page 1954, used in AFML Ch. 17 for event-driven sampling), PELT for offline segmentation, Bayesian Online Change-Point Detection (Adams-MacKay 2007) for streaming. Recent extensions (Tsaknaki-Lillo-Mazzarisi 2024 QF) relax BOCPD's IID-within-regime assumption for financial time series.

Clustering-based regime detection: Two Sigma's publicly documented approach fits a **Gaussian Mixture Model to their 17-dim Factor Lens** and identifies four regimes (Crisis, Steady State, Inflation, Walking on Ice). For a 12-week project, GMM on macro features (VIX, credit spread, term slope, USD, realized correlation) is tractable in 1-2 days and presentable.

Concept-drift detectors for online ML: ADWIN (Bifet-Gavalda 2007) is best for sudden drift; Page-Hinkley better for gradual drift; KSWIN uses KS on sliding windows. Finance-specific work (arXiv 2103.14079) finds DDM/EDDM fail on raw returns; ADWIN on prediction-error stream → trigger retrain is the most reliable pattern.

Conditional signals / factor timing. Cochrane (2011 Presidential Address "Discount Rates") frames the theoretical case that expected returns vary with the business cycle; Lettau-Ludvigson (2001), Bansal-Yaron (2004) provide frameworks. But AQR's "Contrarian Factor Timing is

Deceptively Difficult" (Asness et al. 2017) shows that factor timing *net of existing factor exposures* usually subtracts value — a sobering caveat for any regime-conditional project.

Realistic-in-12-weeks regime methods: HMM (2-3 states) as an *overlay*, not the core signal; GMM on macro features; rolling-window / exponential-decay weighting; fractional differentiation; ADWIN on prediction errors. **Not realistic:** BOCPD with custom likelihoods, deep regime detection (VAE + clustering), full mixture-of-experts per regime, Markov-switching GARCH.

G. Infrastructure and presentation

Internal surface: SecDB objects queryable via Slang or Python-Slang bridges; risk cubes; `gs-quant` module patterns mirror internal APIs (`PricingContext`), `HistoricalPricingContext`, `Dataset`). **Marquee + GS Quant** is the external projection: 400+ datasets, pricing/risk, backtester, Security Master, PlotTool Pro. GS-internal equivalents of JPM's OmniAI exist but are not publicly documented. Standard external tooling: MLflow or Weights & Biases for experiment tracking; `mlfinlab` (Hudson & Thames) for triple-barrier, purged-CV, CPCV, feature importance; `alphalens` for factor evaluation (IC, quantile returns, turnover); `pyfolio` for tear-sheets; `fracdiff` for fractional differentiation; `hmmlearn` / `river` for regime and drift.

Presenting to a trading floor — practitioner-consensus playbook: lead with the causal hypothesis, not the model; report IC / Sharpe / Sortino / turnover / hit rate / drawdown / P&L by regime / capacity, **not** accuracy or F1; one chart per claim; always include a ridge or logistic baseline so the ML lift is visible; quantify transaction-cost sensitivity upfront; state capacity and expected half-life; prefer SHAP for interpretation over raw feature importance; frame ML as *risk management / sizing* (Chan's argument) rather than as "my neural net finds alpha" — traders respond to this framing and it is also defensible.

Part 2 — Project direction proposals

Five concrete directions, ranked roughly by novelty-times-feasibility. All five should share a common infrastructure layer: triple-barrier labels, purged K-fold CV with embargo, transaction-cost-aware backtesting, a ridge baseline on identical features, and Deflated Sharpe + Haircut Sharpe on every reported number.

Project 1 — "Risk as alpha: forced-flow prediction from internal VaR and scenario P&L" (ambitious/novel, highest wow factor). *Pitch:* every external researcher reconstructs intermediary-leverage signals from quarterly Fed Z.1 tables; SecDB has them daily, cross-asset, with correct dealer sign. *What it does:* build daily time series of firm-level Δ VaR, component VaR by asset class, factor-VaR concentration (Herfindahl), stress-scenario P&L rank and dispersion; test whether these predict (a) VIX innovations, (b) drawdowns in the most-concentrated asset class, (c) mean-reversion of crowded factors. *Grounding:* He-Krishnamurthy, Adrian-Etula-Muir, He-Kelly-Manela, Adrian-Brunnermeier CoVaR, Coval-Stafford fire sales. *Data:* SecDB VaR/scenario outputs, factor risk decompositions, book-level Greeks — all internal. *ML:* LightGBM on aggregated risk features + ridge baseline; optionally GMM regime overlay.

Feasibility: tight on 12 weeks but tractable if the intern's desk already runs the risk cubes. **Risk:** internal-use-only; publication / external pitch requires desk sign-off; sample size small for cross-sectional tests. **Wow factor:** highest — directly leverages proprietary data theory predicts should be informative.

Project 2 — "Book-Gamma Intraday Momentum, extended cross-asset" (in-between, highest-probability-of-positive-result). *Pitch:* Baltussen-Da-Lammers-Martens (2021 JFE) show aggregate dealer net gamma predicts last-30-minutes intraday momentum across 60+ futures; extend to non-equity-index markets using real SecDB book Greeks. *What it does:* daily aggregate dealer gamma, vega, vanna, charm across rates futures, G10 FX, credit indices; test whether sign and magnitude predict end-of-day and overnight moves. *ML:* GBM + ridge; panel with asset fixed effects. *Feasibility:* ~6-8 weeks with clean SecDB pulls. *Risk:* publication-adjacent effect may be competed-away on some instruments. *Wow factor:* uses book-level sign (not OI heuristics) → genuinely new evidence.

Project 3 — "Calibration residuals as demand-pressure signals" (novel, defensible). *Pitch:* SVI/SABR/Nelson-Siegel residuals are a byproduct of every nightly calibration and literally cannot be reproduced externally at scale. Gârleanu-Pedersen-Poteshman (2009 RFS) proves they contain demand/flow information. *What it does:* build (date × strike × expiry) residual tensors for the SPX vol surface, a single-name basket, and the USD swaption cube; apply functional PCA; use residual PCs as features in a LightGBM predicting (a) per-strike IV mean-reversion at 1-5 day horizons, (b) short-dated ATM realized vol. *ML:* functional PCA + LightGBM + ridge baseline. *Feasibility:* ~6-8 weeks; hardest part is data plumbing. *Risk:* residuals may be largely liquidity noise rather than information — design experiment to test this cleanly. *Wow factor:* high — a pure "we can do this because we are GS" angle.

Project 4 — "Factor-neutral ML residual strategy on cross-asset Greeks" (safe/conventional, high likelihood of presentable result). *Pitch:* the most defensible design pattern per Gu-Kelly-Xiu / Kelly-Pruitt-Su / Giglio-Kelly-Xiu literature. *What it does:* decompose returns with Barra-style (or IPCA) factors → residuals ε → LightGBM on ε using vol-surface features, cross-asset correlation features, Greeks-aggregated positioning features → decile long-short portfolio with explicit factor-neutrality constraints and transaction costs. *ML:* LightGBM + elastic-net baseline + conditional autoencoder (Gu-Kelly-Xiu 2021) as optional upstream feature extractor. *Feasibility:* highest — 4-6 weeks to a first presentable result. *Risk:* may produce small-cap-concentrated / illiquid signals; the liquidity filter and factor-neutrality constraints kill this if applied too strictly. *Wow factor:* moderate — but it works.

Project 5 — "Arbitrage-constrained Gaussian-process smoothing with uncertainty for an illiquid vol-surface slice" (scoped, niche, high probability of clean deliverable). *Pitch:* long-dated single-name or illiquid-strike vol surfaces have sparse quotes; a no-arbitrage-constrained GP (Cousin-Deleplace-Misko 2022) or deep smoother (Ackerer-Tagasovska-Vatter NeurIPS 2020) provides both interpolation and calibrated uncertainty bands. *What it does:* pick one illiquid corner (e.g., a specific single-name LEAPs grid or long-dated swaption cube), implement GP with arbitrage constraints, benchmark against SVI/SSVI/SABR. *ML:* Gaussian process with monotonicity and convexity constraints; optionally a small NN for comparison. *Feasibility:* ~6-8

weeks, clean scope. *Risk*: narrowest intellectual scope of the five; probably a desk-useful tool rather than an alpha pitch. *Wow factor*: modest, but it is a deliverable a vol desk will use.

Not recommended for a 12-week project: RL for portfolio allocation (FinRL family — fails replication); GNN cross-asset-graph return prediction (leakage-prone literature, no tier-1 public endorsement); deep hedging replication (interesting but narrow; 12 weeks very tight for a credible simulator); "LSTM/Transformer predicts S&P" in any form (almost certain to underperform a GBM and waste weeks).

Part 3 — Annotated bibliography

Essential — read first

López de Prado, *Advances in Financial Machine Learning*, Wiley 2018 (paid book, ~\$60; Reasonable Deviations free notes at https://reasonabledeviations.com/notes/adv_fin_ml/). The validation bible. Triple-barrier labelling, purged/embargoed CV, CPCV, meta-labelling, MDI/MDA/SFI feature importance, fractional differentiation. Essential. Read Chapters 3, 7, 11-12 minimum.

Gu, Kelly, Xiu (2020) "Empirical Asset Pricing via Machine Learning", *Review of Financial Studies* 33(5): 2223-2273. <https://academic.oup.com/rfs/article/33/5/2223/5758276> (paywalled; SSRN preprint free). 60 years of US equity panel, 94 characteristics, ML horse-race. Trees and NNs dominate linear models via nonlinear interactions. The canonical modern reference. Essential.

Kelly, Xiu (2023) "Financial Machine Learning", NBER WP 31502. <https://www.nber.org/papers/w31502> (free). Most comprehensive current survey. Return prediction, factor models, portfolio choice, virtue of complexity, textual data. Essential literature map.

Bailey, López de Prado (2014) "The Deflated Sharpe Ratio", *Journal of Portfolio Management*. <https://www.davidhbailey.com/dhbpapers/deflated-sharpe.pdf> (free). Adjusts reported Sharpe for selection bias, non-normality, sample length. Non-negotiable for any headline number.

Bailey, Borwein, López de Prado, Zhu (2014) "Pseudo-Mathematics and Financial Charlatanism", *Notices of the AMS*. <https://www.davidhbailey.com/dhbpapers/backtest-pseudo.pdf> (free). Minimum Backtest Length formula. Essential.

Harvey, Liu (2015) "Backtesting", *Journal of Portfolio Management*. PDF at https://people.duke.edu/~charvey/Research/Published_Papers/P120_Backtesting.PDF; code at <https://people.duke.edu/~charvey/backtesting/> (free). Haircut Sharpe methodology with Bonferroni/Holm/BHY adjustments. Essential.

Harvey, Liu, Zhu (2016) "...and the Cross-Section of Expected Returns", *RFS* 29(1). <https://academic.oup.com/rfs/article-abstract/29/1/5/1843824>. 316+ factors; $t > 3$ hurdle for new factors. Essential for thinking about feature-search discipline.

GS Developer Blog "SecDb Observability Journey" (Oct 2023).

<https://developer.gs.com/blog/posts/secdb-observability-journey> (free). Only first-party GS post detailing current SecDB architecture. Essential context.

gs-quant **GitHub**. <https://github.com/goldmansachs/gs-quant> (free, Apache 2.0). Closest public proxy for SecDB data model; practice with this. Essential for pre-internship preparation.

Recommended — high value

López de Prado (2020) *Machine Learning for Asset Managers*, Cambridge Elements.

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3558728 (~\$25). Denser companion to AFML. False Strategy Theorem, clustered MDI, HRP, denoising covariance matrices. Recommended.

Zeng, Chen, Zhang, Xu (2023) "Are Transformers Effective for Time Series Forecasting?", AAAI. <https://arxiv.org/abs/2205.13504> (free). DLinear beats every tested transformer on 9/9 benchmarks. Recommended.

Gu, Kelly, Xiu (2021) "Autoencoder Asset Pricing Models", *J. Econometrics* 222(1). Cleanest evidence autoencoders add value when embedded in a no-arbitrage factor structure. Recommended.

Kozak, Nagel, Santosh (2020) "Shrinking the Cross-Section", *JFE* 135(2).

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2945663 (free). Ridge-shrunk SDF on PC characteristics matches non-linear ML. Makes the ridge baseline case. Recommended.

Kelly, Pruitt, Su (2019) "Characteristics are Covariances", *JFE* 134(3) (IPCA).

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3032013 (free preprint). Latent factors with characteristic-instrumented loadings; extended to options and bonds. Recommended cross-asset framework.

He, Krishnamurthy (2013) "Intermediary Asset Pricing", *AER* 103(2).

<https://www.aeaweb.org/articles?id=10.1257/aer.103.2.732>. Theoretical foundation for risk-as-alpha. Recommended.

Adrian, Etula, Muir (2014) "Financial Intermediaries and the Cross-Section of Asset Returns", *J. Finance*.

https://www.newyorkfed.org/medialibrary/media/research/staff_reports/sr464.pdf (free preprint). $R^2=77\%$ single-factor SDF from broker-dealer leverage. Recommended.

He, Kelly, Manela (2017) "Intermediary Asset Pricing: New Evidence from Many Asset Classes", *JFE*. <https://www.sciencedirect.com/science/article/abs/pii/S0304405X1730212X>.

Single kernel across equity/options/CDS/bonds/FX/commodities. Data on Manela's website. Recommended.

Adrian, Shin (2010) "Liquidity and Leverage", *JFI*.

https://www.newyorkfed.org/medialibrary/media/research/staff_reports/sr328.pdf. Dealer repos forecast VIX innovations — direct evidence for risk-as-alpha. Recommended.

Baltussen, Da, Lammers, Martens (2021) "Hedging Demand and Market Intraday Momentum", *JFE*. <https://www3.nd.edu/~zda/intramom.pdf> (free). Dealer gamma determines last-30-min intraday momentum across 60+ futures. The direct template for Project 2. Recommended.

Barbon, Buraschi (2021) "Gamma Fragility". SSRN 3725454 (free). Stock-level gamma imbalance $\times$ illiquidity drives intraday momentum vs reversal. Recommended.

Xing, Zhang, Zhao (2010) "What Does the Individual Option Volatility Smirk Tell Us?", *JFQA*. $\sim 10.9\%/yr$ cross-sectional smirk signal. Recommended.

Cremers, Weinbaum (2010) "Deviations from Put-Call Parity and Stock Return Predictability", *JFQA*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=968237. +50bps/week IV-spread signal. Recommended.

Bollerslev, Tauchen, Zhou (2009) "Expected Stock Returns and Variance Risk Premia", *RFS*. VRP predicts SPX at quarterly horizon. Recommended.

Gârleanu, Pedersen, Poteshman (2009) "Demand-Based Option Pricing", *RFS*. Theoretical foundation for calibration-residuals-as-signals (Project 3). Recommended.

Muravyev, Pearson, Pollet (2022) showing IVS and skew partly proxy for borrow fees. Critical caveat on option-implied equity signals — always control for short interest. Recommended.

Cont, da Fonseca (2002) "Dynamics of Implied Volatility Surfaces", *Quantitative Finance*. 3-factor PCA of daily IV surfaces. Foundation for surface features. Recommended.

Du, Tepper, Verdelhan (2018) "Deviations from Covered Interest Rate Parity", *J. Finance*. Cross-currency basis as dealer balance-sheet stress proxy. Recommended.

Koijen, Moskowitz, Pedersen, Vrugt (2018) "Carry", *JFE*. <https://spinup-000d1a-wp-offload-media.s3.amazonaws.com/faculty/wp-content/uploads/sites/3/2019/04/Carry.pdf> (free). Unified carry across 9 asset classes. Recommended for cross-asset framing.

Asness, Moskowitz, Pedersen (2013) "Value and Momentum Everywhere", *J. Finance*. Data at aqr.com/Insights/Datasets. Canonical cross-asset factor reference. Recommended.

Moskowitz, Ooi, Pedersen (2012) "Time Series Momentum", *JFE*. <http://docs.lhpedersen.com/TimeSeriesMomentum.pdf> (free). 58 futures panel. Recommended.

Two Sigma "A Machine Learning Approach to Regime Modeling". <https://www.twosigma.com/articles/a-machine-learning-approach-to-regime-modeling/> (free). GMM on 17-dim factor lens $\rightarrow$ 4 regimes. The practitioner template for Project's regime overlay. Recommended.

Adrian, Brunnermeier (2016) "CoVaR", *AER*.

<https://markus.scholar.princeton.edu/document/188>. Forward-CoVaR as leading systemic-risk signal. Recommended.

Coval, Stafford (2007) "Asset Fire Sales and Purchases in Equity Markets", *JFE*. Mutual-fund forced selling predicts 5-day price pressure with reversals. Recommended foundation for fire-sale mechanics.

Bailey, Borwein, López de Prado, Zhu (2016) "Probability of Backtest Overfitting", *J. Computational Finance*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2326253. CSCV for PBO. Recommended companion to DSR.

Asness, Chandra, Iltis, Israel (2017) "Contrarian Factor Timing is Deceptively Difficult", *J. Portfolio Management*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2928945. Factor timing net of existing exposures often subtracts value. Recommended caveat on regime-conditional signals.

Carver (2015) *Systematic Trading*, Harriman House / blog at qoppac.blogspot.com. Anti-overfitting discipline; three types of fitting. Recommended for research-process sanity.

Carver SlideShare "Trading Strategies That Are Designed Not Fitted". <https://www.slideshare.net/slideshow/trading-strategies-that-are-designed-not-fittedby-robert-carver-independent-systematic-futures-trader-writer-and-research-consultant/75640564> (free). 45-slide synthesis of the idea-first method. Recommended.

Jansen (2020) *Machine Learning for Algorithmic Trading*, 2nd ed, Packt. <https://github.com/stefan-jansen/machine-learning-for-trading> (code free). Best end-to-end Python reference for pipeline boilerplate. Recommended.

mlfinlab (Hudson & Thames). <https://github.com/hudson-and-thames/mlfinlab>. Production-ready triple-barrier, purged CV, CPCV. Double-check on your data; has had bugs. Recommended tooling.

Optional — skim for ideas

Nie et al. (2023) "A Time Series Is Worth 64 Words" (PatchTST), ICLR. <https://arxiv.org/abs/2211.14730> (free). Current credible transformer baseline if deep time-series attempted. Optional.

Challu et al. (2022) N-HiTS. Strong MLP baseline for forecasting. Optional.

Buehler, Gonon, Teichmann, Wood (2019) "Deep Hedging", *Quantitative Finance*. <https://arxiv.org/abs/1802.03042>. RL hedging under frictions. Optional (scope stretch only).

Nevmyvaka, Feng, Kearns (2006) "RL for Optimized Trade Execution", ICML. <https://www.cis.upenn.edu/~mkearns/papers/rlexec.pdf>. Foundational RL execution paper. Optional.

Ackerer, Tagasovska, Vatter (2020) "Deep Smoothing of the Implied Volatility Surface", *NeurIPS*. Direct reference for Project 5. Optional.

Cousin, Deleplace, Misko (2022) GP with arbitrage constraints for swaption cube, *Risks*. Direct reference for Project 5. Optional.

Feng, Giglio, Xiu (2020) "Taming the Factor Zoo", *J. Finance*. Factor-selection methodology. Optional.

Kelly, Palhares, Pruitt (2023) corporate-bond characteristics model, *J. Finance*. Optional for credit-focused extensions.

Chen, Pelger, Zhu (2024) "Deep Learning in Asset Pricing", *Management Science*. <https://arxiv.org/abs/1904.00745>. No-arbitrage-constrained GAN+LSTM; OOS Sharpe ~2.1. Optional.

Freyberger, Neuhierl, Weber (2020) "Dissecting Characteristics Nonparametrically", *RFS*. Adaptive group LASSO. Optional.

Ofek, Richardson, Whitelaw (2004) parity-violations paper, *JFE*. Optional complement to Cremers-Weinbaum.

Pan, Poteshman (2006) "Information in Option Volume", *RFS*. Optional for option-volume signals.

Lustig, Roussanov, Verdelhan (2011) "Common Risk Factors in Currency Markets", *RFS*. https://www.nber.org/system/files/working_papers/w14082/w14082.pdf. FX carry factor model. Optional.

Acharya, Johnson (2007) CDS-equity lead-lag, *JFE*. Optional for credit→equity.

Hilscher, Pollet, Wilson (2015) counter-evidence, *JFQA*. Optional balance to Acharya-Johnson.

Gorton, Hayashi, Rouwenhorst (2012) "Fundamentals of Commodity Futures Returns". Optional for commodity basis signals.

Laloux, Cizeau, Bouchaud, Potters (1999) "Noise Dressing of Financial Correlation Matrices". Optional for RMT eigenvalue features.

Kritzman, Li, Page, Rigobon (2011) "Absorption Ratio", *Financial Analysts Journal*. Optional systemic-risk feature.

Adams, MacKay (2007) BOCPD. <https://arxiv.org/abs/0710.3742>. Optional for advanced regime work.

Bifet, Gavalda (2007) ADWIN. Optional for online-learning drift detection.

Cochrane (2011) "Discount Rates", *J. Finance*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1820084. Optional theoretical framing for conditional signals.

Ilmanen *Expected Returns* (2011), *Investing Amid Low Expected Returns* (2022). Optional big-picture grounding.

Bennett "Trading Volatility" (free PDF, trading-volatility.com). Optional practitioner primer on dispersion, skew, variance swaps.

Gatheral *The Volatility Surface* (2006). Optional canonical vol-surface reference.

Bergomi *Stochastic Volatility Modeling* (2016). Optional for forward-variance models and SSR.

Sinclair *Volatility Trading / Positional Option Trading*. Optional practitioner on VRP harvesting.

Illing, Aaron (2006) Bank of Canada survey of risk-appetite indices.

<https://www.bankofcanada.ca/wp-content/uploads/2012/01/fsr-0605-illing.pdf>. Best public description of GSRAI/JPM LCPI/CSFB GRAI construction. Optional.

Kolanovic, Wei JPM Systematic Cross-Asset Strategy reports (2013, 2015).

<https://www.cmegroup.com/education/files/jpm-systematic-strategies-2013-12-11-1277971.pdf>. Optional practitioner reference.

QuantStart HMM tutorial. <https://www.quantstart.com/articles/hidden-markov-models-for-regime-detection-using-r/>. Optional practical walkthrough.

Martin Chavez HBS Module 2. <https://www.rmartinchavez.com/module2> (free). Optional SecDB design-intent backstory.

Michal Dorko InfoQ/QCon talk on Slang. <https://www.infoq.com/presentations/slang/> (free). Optional Slang internals.

PyData London 2018 "Python at Massive Scale" (JPM Athena).

<https://www.youtube.com/watch?v=ZYD9yyMh9Hk> (free). Optional analogous-system reference.

Ernie Chan blog (epchan.com) and "Backtesting and Its Pitfalls" PDF.

<https://epchan.com/img/links/Backtesting-and-its-Pitfalls.pdf>. Optional practitioner wisdom.

Harvey, Hoyle, Rattray et al. (2019) "The Best of Strategies for the Worst of Times", *JPM*. Optional crisis-alpha reference.

Quantocracy aggregator. <https://quantocracy.com/>. Optional — daily quant-blog feed for ongoing discovery.

Skim-for-ideas — treat critically

Feng, He, Wang, Luo, Liu, Chua (2019) "Temporal Relational Ranking for Stock Prediction" (RSR), ACM TOIS. <https://arxiv.org/abs/1809.09441>. Representative GNN paper. Skim for architecture; discount backtest numbers (likely leakage).

Kim et al. (2019) HATS. <https://arxiv.org/abs/1908.07999>. Skim for architecture only.

FinRL and RL-portfolio-allocation arXiv preprints. Skim critically — most fail replication.

SpotGamma / Tier1Alpha / MenthorQ blogs on dealer gamma. Useful intuition-building on dealer-flow mechanics; do not cite as rigorous evidence. Public GEX assumes dealer side which is ~30% wrong.

Gaps and honest unknowns

The "SecDB originators" in the task prompt ("Armstrong/Korn") do not match any public source; documented originators are Dubno, Gribble, Lundeen under Avanessians. Confirm with the internship sponsor. **Public information on SecDB's internal schema, precise risk-measure availability per asset class, intraday vs end-of-day granularity of risk cubes, and the name of GS's internal ML research platform runs out quickly** — the intern's sponsor will know; public researchers do not. **Morgan Stanley and Deutsche Bank's core risk platforms are essentially undocumented publicly.** Renaissance, Citadel, DE Shaw production methodologies are unavailable; all statements about what tier-1 firms "actually do" are inferences from competition performance, hiring, and sparse public writings. **Risk-neutral-skewness-return sign is genuinely contested** in the academic literature (Conrad-Dittmar-Ghysels negative, Stilger-Kostakis-Poon positive, term-dependent per Feunou et al.); treat any RNS-based feature as conditional. **Muravyev-Pearson-Pollet (2022) critique** — IVS and skew signals may largely proxy for stock-borrow fees rather than information — is important and understudied in ML contexts; any project using these signals must control for short interest and borrow cost. **"Virtue of complexity"** (Kelly-Malamud-Zhou 2022; Didisheim 2023) is theoretically elegant but empirically contested. **Copper-gold ratio and various practitioner regime-gating heuristics** have broken since 2021 and should not be relied on. **Published signals decay 30-50% post-publication** (McLean-Pontiff 2016) — internal GS signals may have decayed similarly, and any "rediscovery" of a known academic signal in SecDB should be viewed skeptically.

Raw source log

Primary SecDB/Slang: GS Developer Blog SecDb Observability Journey; R. Martin Chavez rmartinchavez.com Module 2; a16z In the Vault with Chavez; InfoQ Dorko Slang talk; eFinancialCareers SecDB articles 2017 and 2023; Wikipedia Michael Dubno; HBS Case 518-039; Risk.net Beacon 2017; Wharton FinTech Mark Higgins interview; PyGotham 2018 BofA 17M-lines-of-Python talk; PyData London 2018 JPM Athena talk; TechRepublic Athena 35M-lines article.

GS Marquee / GS Quant: marquee.gs.com/welcome/our-platform/gs-quant; developer.gs.com/docs; github.com/goldmansachs/gs-quant; pypi.org/project/gs-quant.

ML foundational: Gu-Kelly-Xiu 2020 RFS; Kelly-Xiu 2023 NBER 31502; Gu-Kelly-Xiu 2021 JEcon; Kozak-Nagel-Santosh 2020 JFE; Kelly-Pruitt-Su 2019 JFE; Freyberger-Neuhierl-Weber 2020 RFS; Chen-Pelger-Zhu 2024 MgmtSci arXiv 1904.00745; Feng-Giglio-Xiu 2020 JF; Chinco-Clark-

Joseph-Ye 2019 JF; Israel-Kelly-Moskowitz 2020 JoIM; Zeng et al. 2023 AAAI arXiv 2205.13504; Nie et al. 2023 ICLR arXiv 2211.14730; Challu et al. 2022 N-HiTS; Feng et al. 2019 RSR arXiv 1809.09441; Kim et al. 2019 HATS arXiv 1908.07999; Nevmyvaka-Feng-Kearns 2006 ICML; Buehler et al. 2019 QF arXiv 1802.03042; Kolm-Ritter 2019 JFDS; Hull-Cao 2020.

Validation: López de Prado AFML 2018, MLAM 2020; Bailey-Borwein-LdP Notices AMS 2014; Bailey-LdP DSR 2014 SSRN 2460551; Bailey et al. PBO 2016 SSRN 2326253; Harvey-Liu Backtesting JPM 2015; Harvey-Liu-Zhu 2016 RFS; Chen 2024 arXiv 2206.15365; mfinlab hudsonthames; pypbo; Carver SlideShare and qoppac.blogspot; Chan epchan.com and Backtesting-and-its-Pitfalls PDF; Jansen 2020 Packt + GitHub; Quantocracy.

Feature engineering (options/vols): Barbon-Buraschi SSRN 3725454; Ni-Pearson-Potesman RFS 2021 SSRN 2867461 and JFE 2005 SSRN 519044; Baltussen-van Bekkum-van der Grient JFQA 2018 SSRN 2023066; Baltussen-Da-Lammers-Martens JFE 2021; Gârleanu-Pedersen-Potesman RFS 2009; Xing-Zhang-Zhao JFQA 2010 SSRN 1107464; Cremers-Weinbaum JFQA 2010 SSRN 968237; Bali-Hovakimian MgmtSci 2009 SSRN 1029197; An-Ang-Bali-Cakici JF 2014 SSRN 1533089; Bollerslev-Tauchen-Zhou RFS 2009 SSRN 948309; Conrad-Dittmar-Ghysels JF 2013 SSRN 1522218; Stilger-Kostakis-Poon MgmtSci 2017; Bali-Hu-Murray 2017; Muravyev-Pearson-Pollet 2022; Cont-da Fonseca QF 2002 SSRN 295859; Avellaneda et al. 2020 arXiv 2002.00085; Bergomi Smile Dynamics SSRN 1493302; Driessen-Maenhout-Vilkov JF 2009; Laloux-Cizeau-Bouchaud-Potters PRL 1999; Bouchaud-Potters 2009 arXiv 0910.1205; Akerer-Tagasovska-Vatter NeurIPS 2020; Cousin-Deleplace-Misko Risks 2022; Gatheral 2006 and Bergomi 2016 books; Bennett Trading Volatility PDF.

Rates/credit/cross-asset: Diebold-Li 2006 JEcon; Du-Tepper-Verdelhan JF 2018; Bai-Collin-Dufresne JFE 2019; Acharya-Johnson JFE 2007; Hilscher-Pollet-Wilson JFQA 2015; Cao-Yu-Zhong JFM 2010; Gorton-Hayashi-Rouwenhorst 2012; Lustig-Roussanov-Verdelhan RFS 2011 NBER 14082; Koijen-Moskowitz-Pedersen-Vrugt JFE 2018; Brunnermeier-Nagel-Pedersen 2008; Moskowitz-Ooi-Pedersen JFE 2012; Asness-Moskowitz-Pedersen JF 2013.

Intermediary/risk-as-alpha: He-Krishnamurthy AER 2013; Adrian-Etula-Muir JF 2014 NYFed SR464; He-Kelly-Manela JFE 2017; Adrian-Shin JFI 2010 NYFed SR328; Adrian-Moench-Shin SSRN 2321180; Adrian-Brunnermeier AER 2016; Coval-Stafford JFE 2007; Etula 2013; Kritzman-Li-Page-Rigobon FAJ 2011; Illing-Aaron Bank of Canada 2006; Kolanovic JPM cmegroup.com 2013 and 2015.

Regime/drift: Hamilton Econometrica 1989; Adams-MacKay 2007 arXiv 0710.3742; Tsaknaki-Lillo-Mazzarisi QF 2024; Bifet-Gavaldà 2007 ADWIN; Gama et al. 2004 DDM; arXiv 2103.14079; AUA Capstone 2025; QuantStart HMM tutorial; LSEG regime article; arXiv 2310.04536; Two Sigma regime modeling article; Man Group Harvey et al. 2019 JPM; Asness-Chandra-Ilmanen-Israel JPM 2017 SSRN 2928945; Cochrane JF 2011 SSRN 1820084; Ilmanen 2011 and 2022.